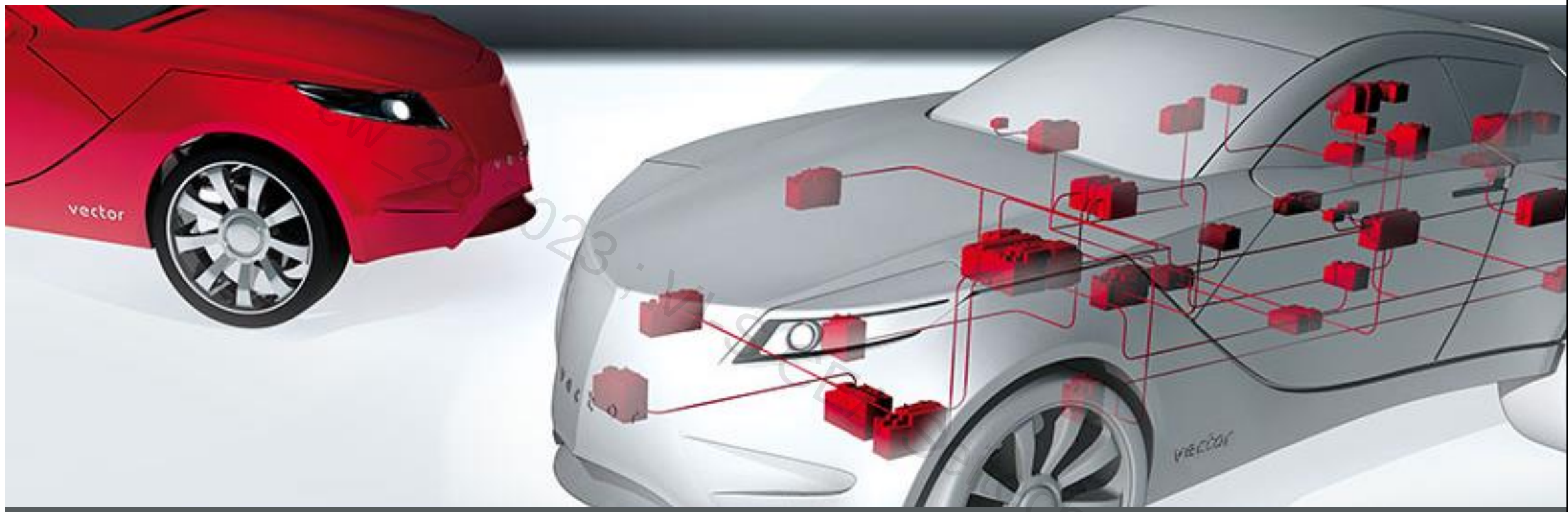
VECTOR



AUTOSAR in Practice

Introduction

Agenda

- **Introduction**
- Workflows
- Coming up next

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Prerequisite for this training

1 day

AUTOSAR Seminar

Basics and Theory

3 days

AUTOSAR in Practice

Practice and Experience

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How does this training work?

e.g.

Mode Management

Training Slides

- Agenda
- Introduction
- Operating System
- Software Components
- I/O
- Communication
- Mode Management
- Memory
- Diagnostics
- Busses (for reference only)

Slides are divided into

- Theory
- Exercise description

THEORY	EXERCISE
Theory	Exercise Description

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Recap: What we already know

…from AUTOSAR fundamental training or from other sources

- Software Components (SWCs)
  - Compositions
  - Atomics
- Architecture
  - Hierarchy of SWCs
- Methodology
  - Workflow between OEM and Tier 1
- Exchange Data Formats
  - ARXML schema
  - ECUEX, ECUC, BSWMD, and SWC descriptions
- RTE
  - Middleware which makes use of OS and BSW and abstracts communication
- BSW
  - Functional clusters, layered SW Architecture
- Tooling has a vital role
  - Work can only be accomplished with AUTOSAR Tools

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VECTOR

What is coming up in the next three days?

- Acquire working experience with the DaVinci Tools (Developer, Configurator Pro)
  - Configure the BSW with Configurator Pro
  - Set up Software Components (SWCs)
  - Assemble SWCs in a software design
  - Generate embedded C code for an ECU
  - Generate and debug the RTE
  - Program Runnables
  - Simulate the result via CANoe

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Let’s start >>

Agenda

Introduction

- **Workflows**

Coming up next

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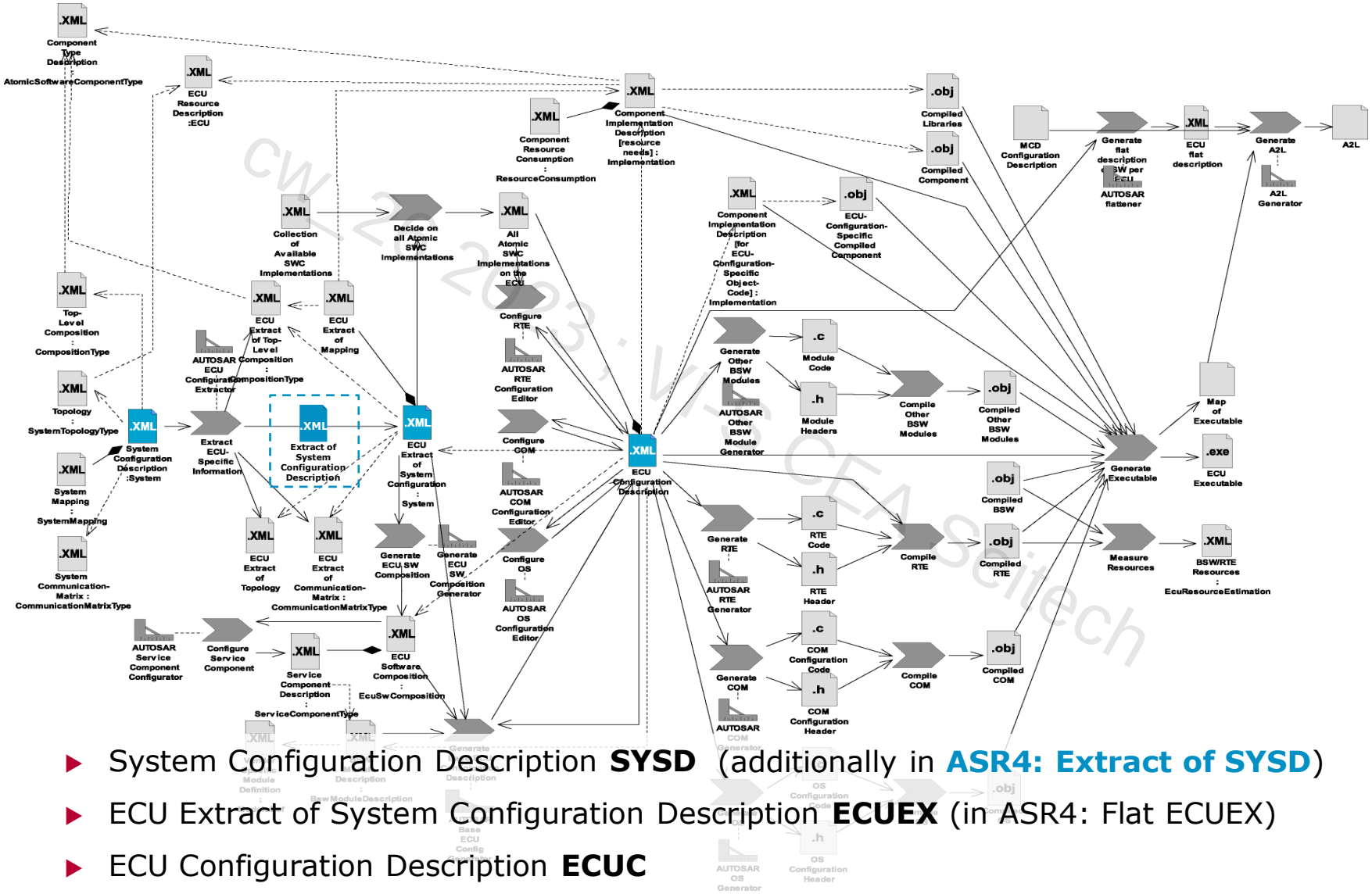
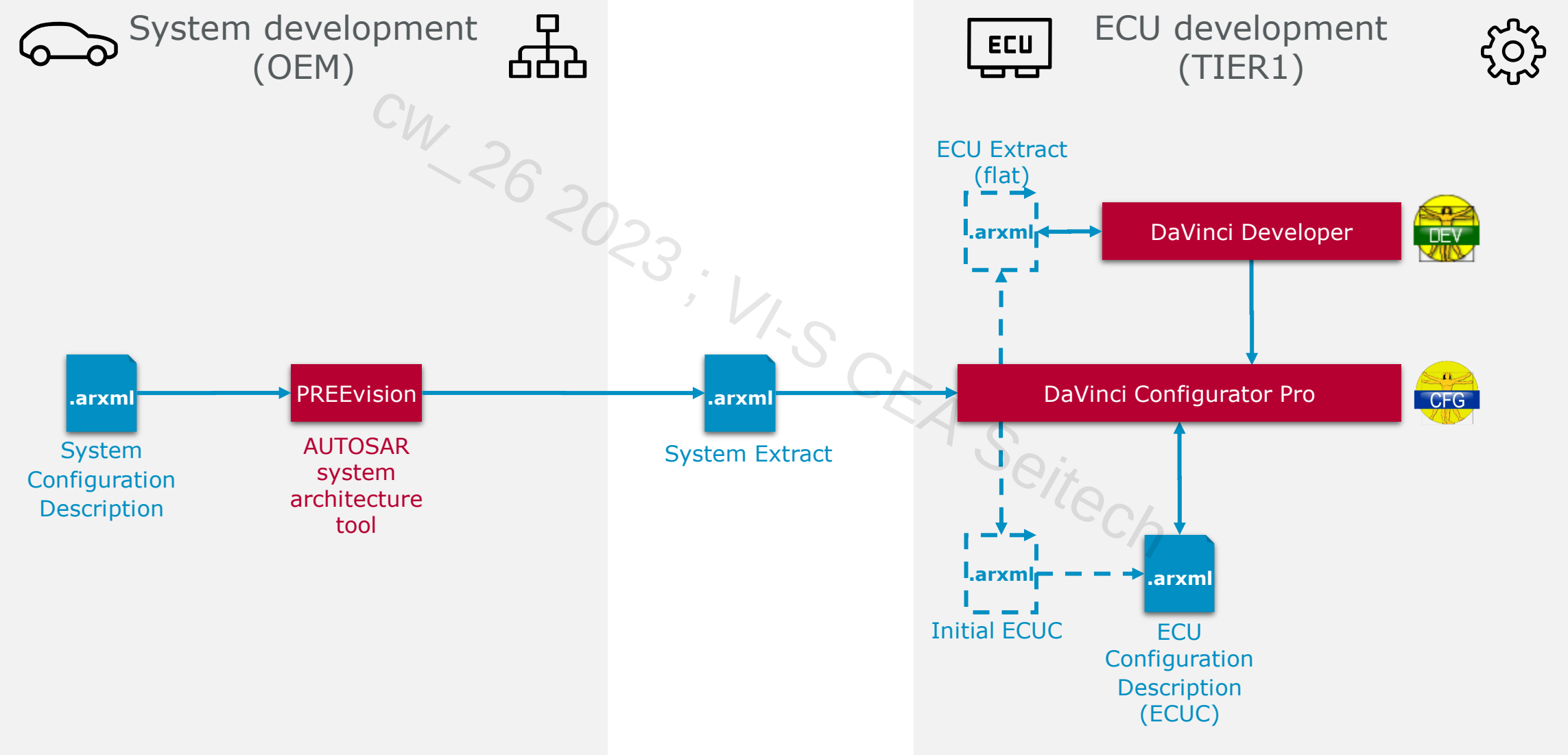
The AUTOSAR Methodology



- System Configuration Description **SYSD** (additionally in **ASR4: Extract of SYSD**)
- ECU Extract of System Configuration Description **ECUEX** (in ASR4: Flat ECUEX)
- ECU Configuration Description **ECUC**

Division of work between OEM and TIER1



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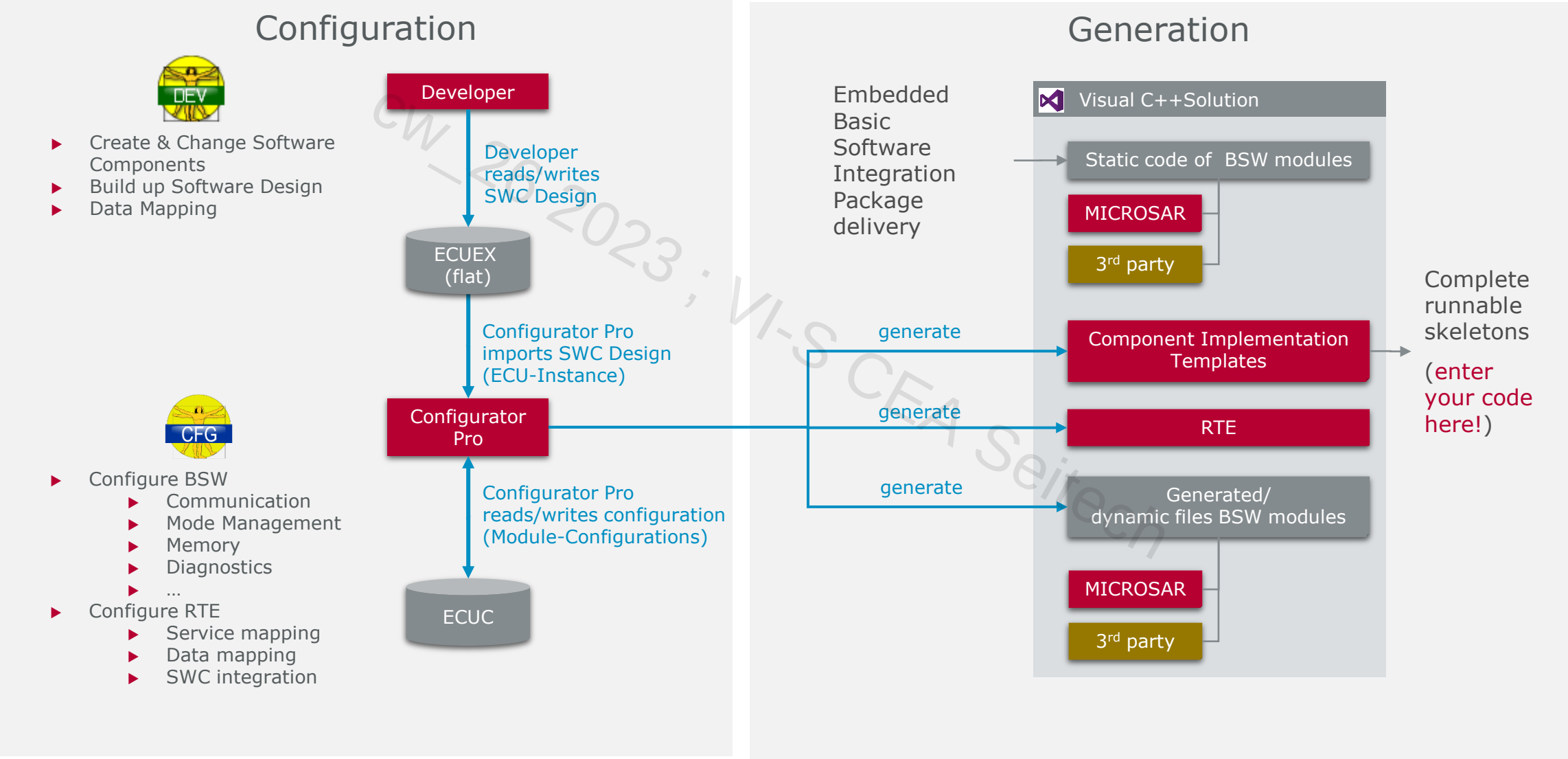
ECU configuration and generation of configuration dependent code



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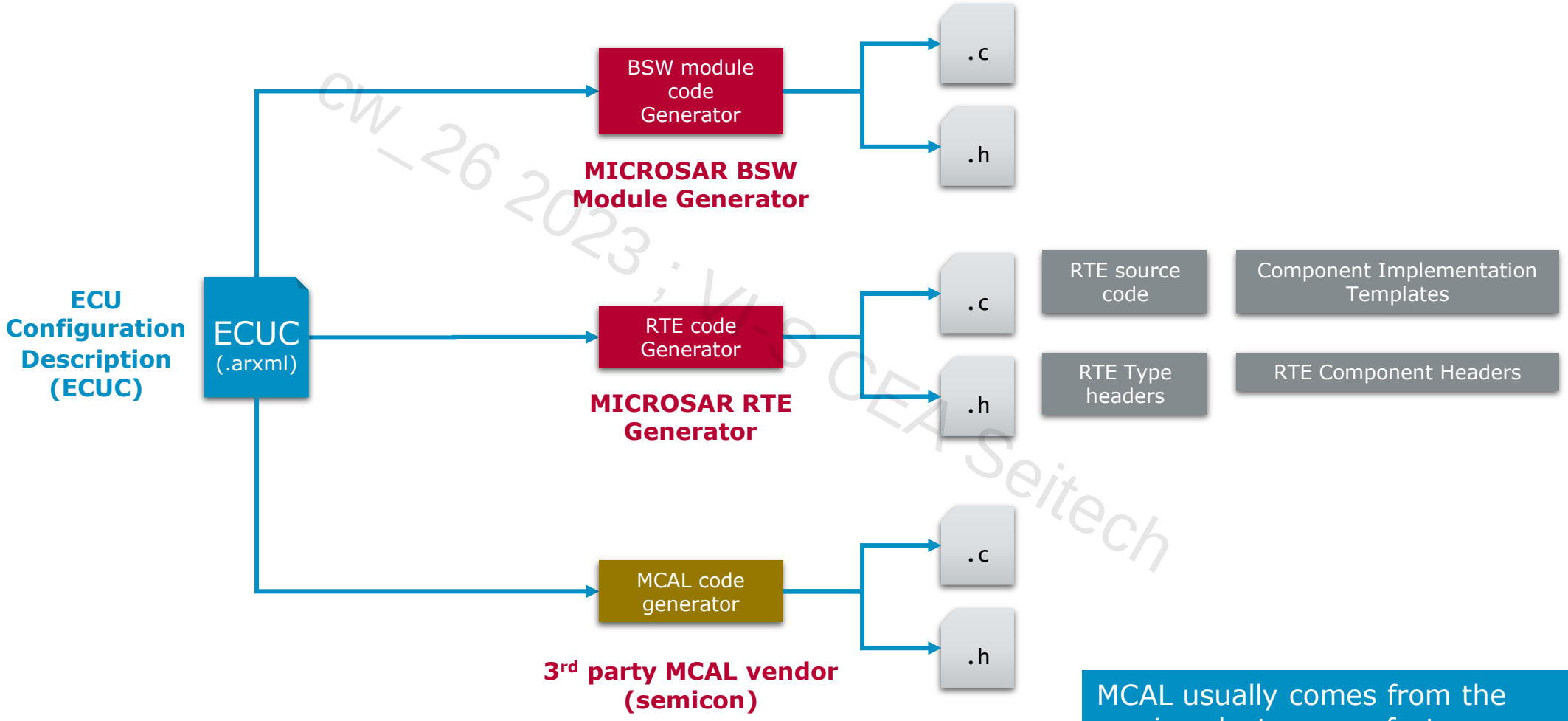
Embedded Software Code Generation



MCAL usually comes from the semiconductor manufacturer
exception: communication drivers

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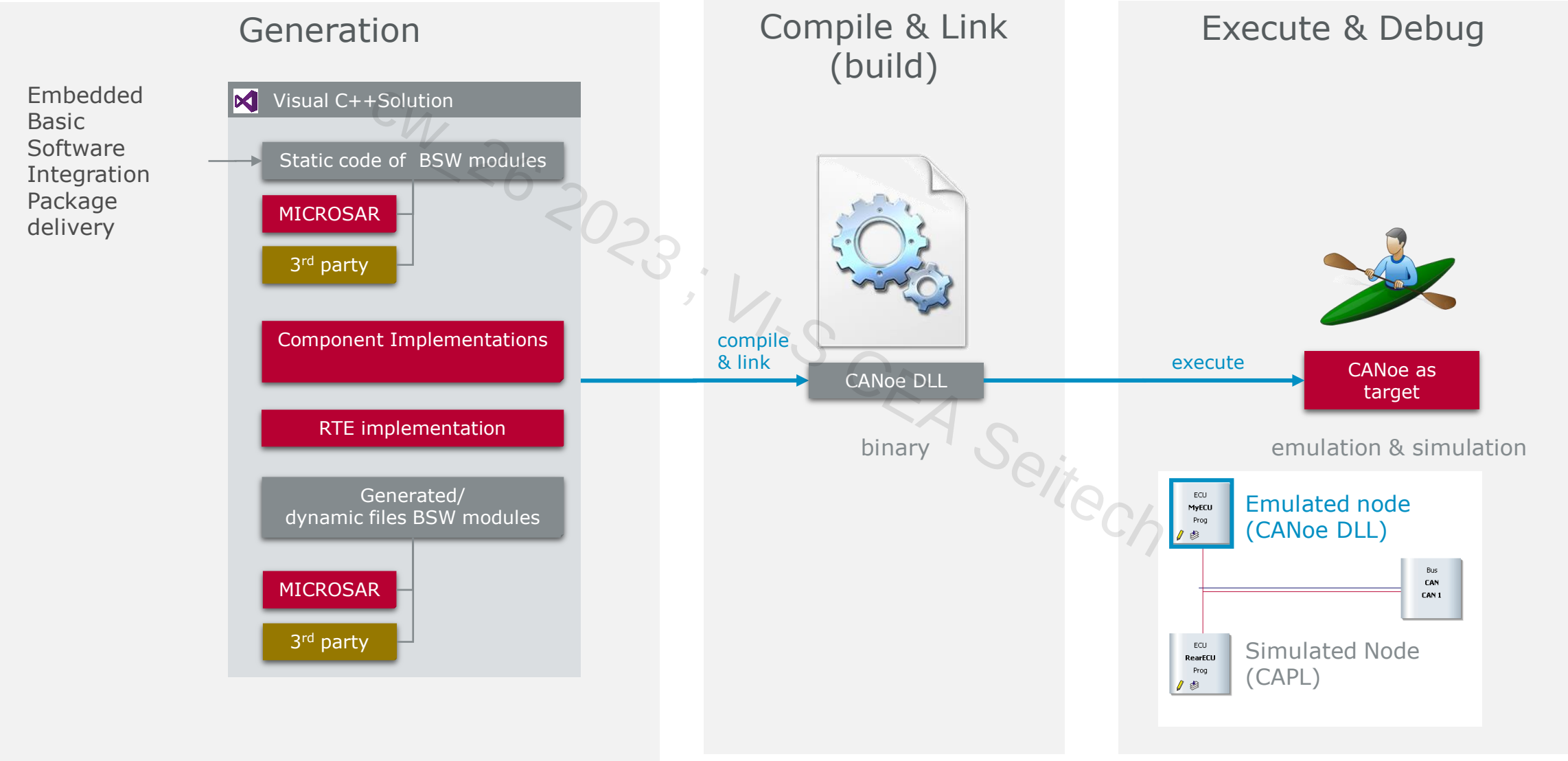
build, execute and debug



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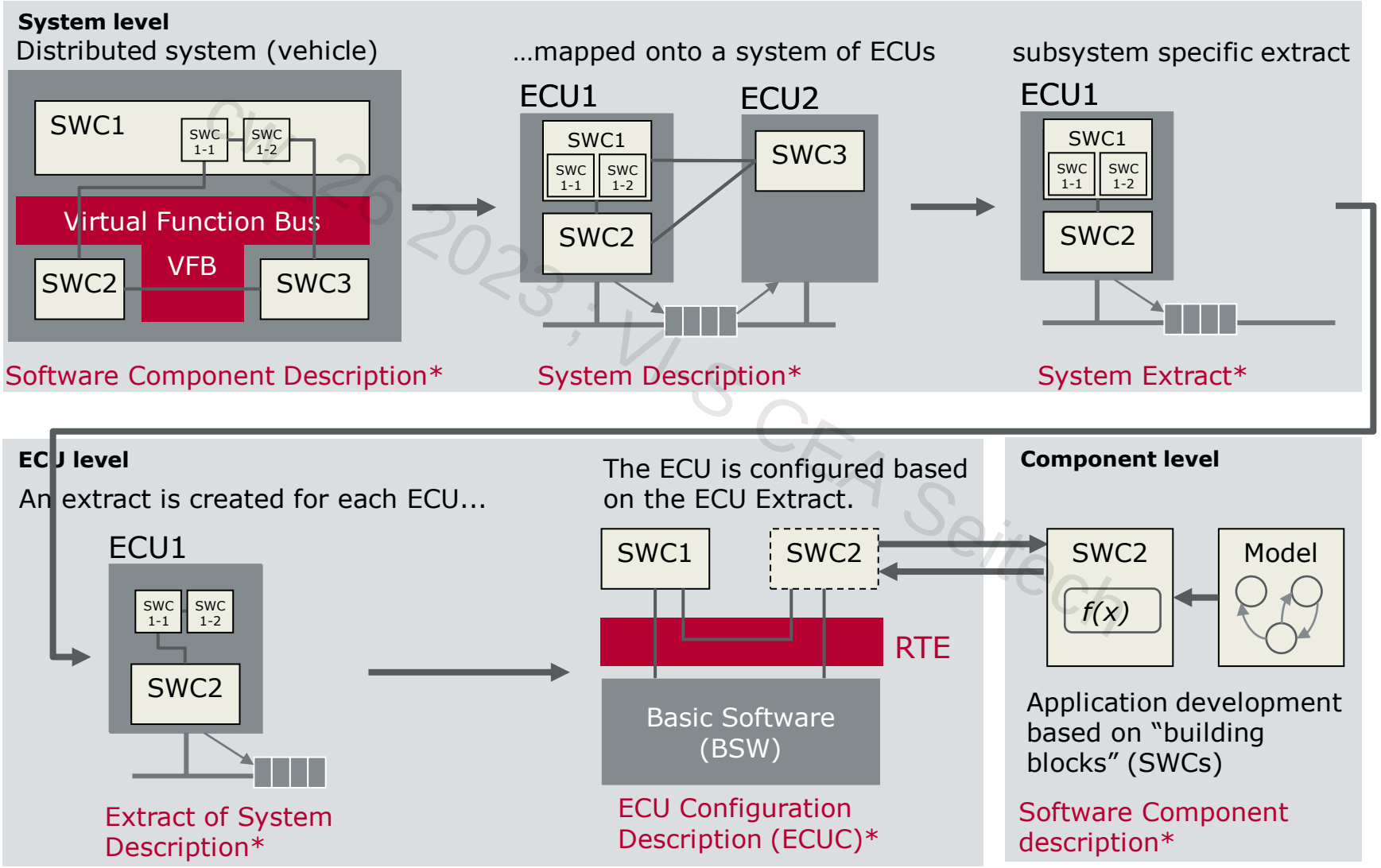
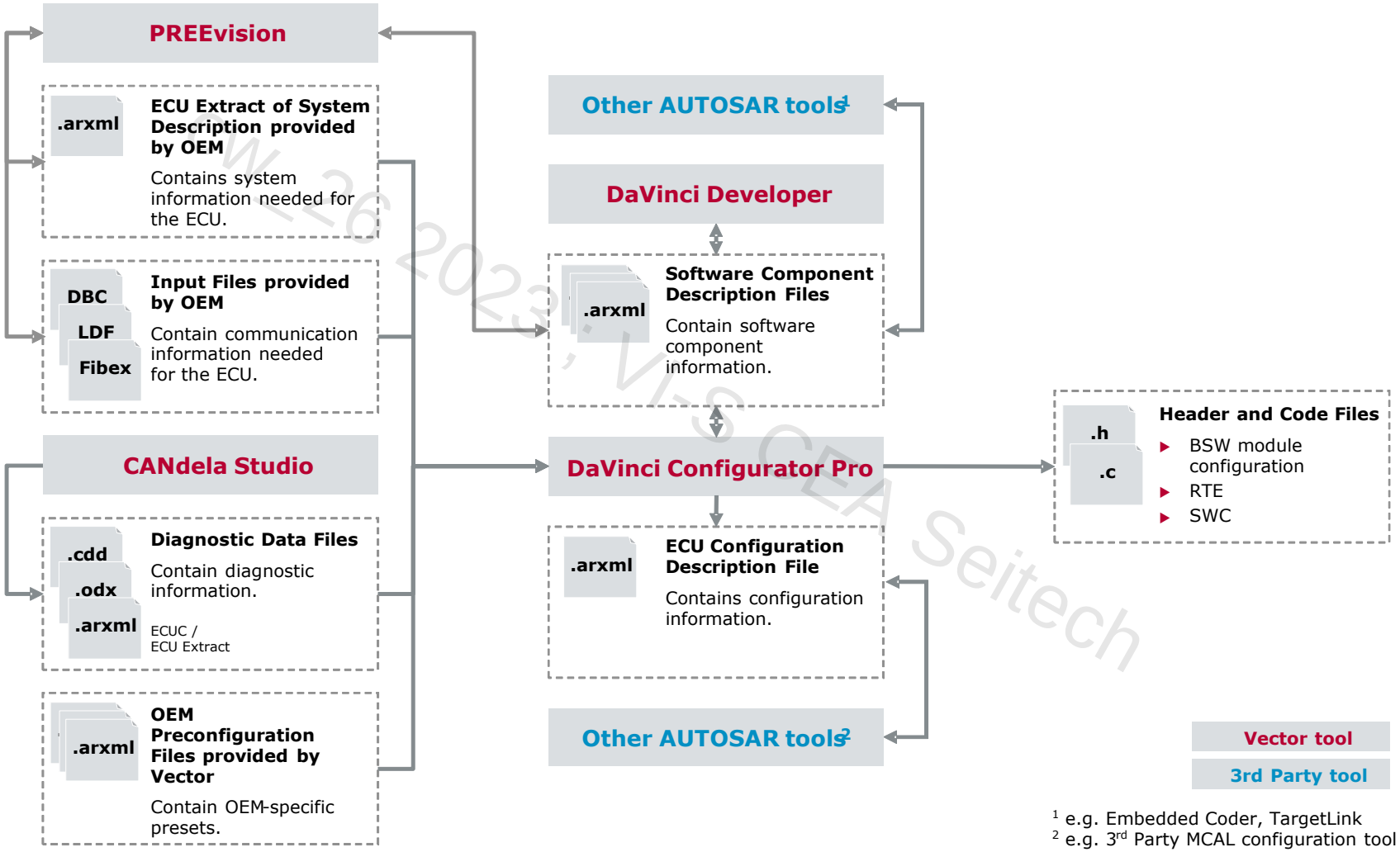
AUTOSAR methodology in a nutshell



*) AUTOSAR

Complete MICROSAR Workflow according to AUTOSAR 4.x



[1] e.g. Embedded Coder, TargetLink
[2] e.g. 3rd Party MCAL configuration tool

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Introduction

Operating System

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